1 n is an integer such that $-2 \leq n < 3$
Write down all the possible values of n.

-2, -1, 0, 1, 2

(Total for question 1 is 2 marks)

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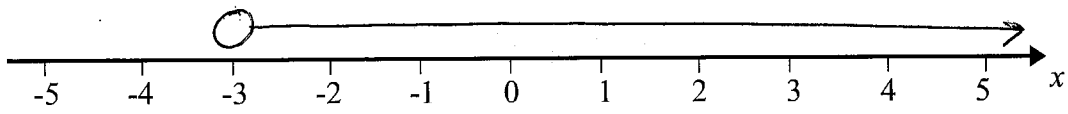
2 (a) On the number line, show the inequality $x > -3$



(2)

$1 \leq y < 5$ where y is an integer.

(b) Write down all the possible values of y.

1, 2, 3, 4

(2)

(c) Solve $4t + 7 \leq 19$

$-7 \quad -7$

$4t \leq 12$

$t \leq 3$

$t \leq 3$

(2)

(Total for question 2 is 6 marks)

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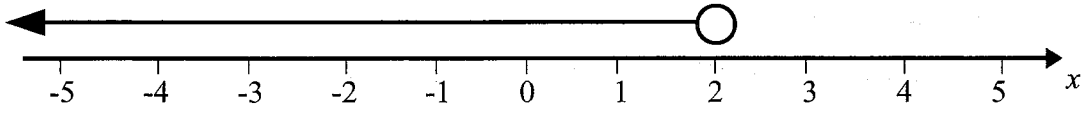
3 Write down the inequality shown on the number line.



$x < 2$

(Total for question 3 is 2 marks)

4 (a) $-1 < n \leq 3$ where n is an integer.

(b) Write down all the possible values of n.

0, 1, 2, 3

(2)

(c) Solve $2x - 5 > 8$

$+5 \quad +5$

$2x > 13$

$x > \frac{13}{2}$

$x > \frac{13}{2}$

(2)

(Total for question 4 is 4 marks)

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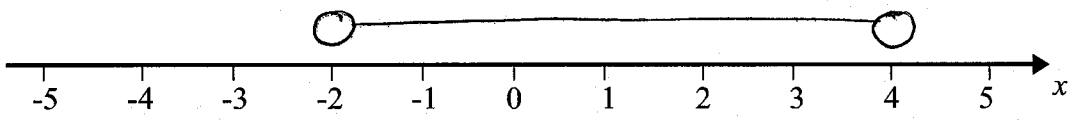
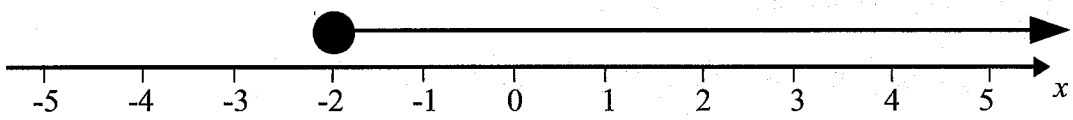
5 (a) On the number line, show the inequality $-2 < x < 4$



(2)

(b) Write down the inequality shown on the number line.

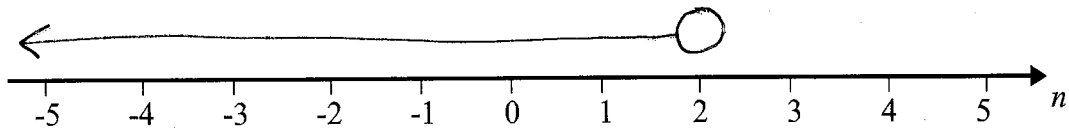


$x \geqslant -2$

(2)

(Total for question 5 is 4 marks)

6 (a) On the number line, show the inequality $n < 2$.



(2)

$4 \leq y < 8$ where y is an integer.

(b) Write down all the possible values of y.

4, 5, 6, 7

(2)

(c) Solve $4x + 6 \leq x + 21$

$-x \qquad -x$

$3x + 6 \leq 21$

$-6 \qquad -6$

$3x \leq 15$

$x \leq 5$

$x \leq 5$

(3)

(Total for question 6 is 7 marks)

---

7 Solve $4x \leq x + 6$

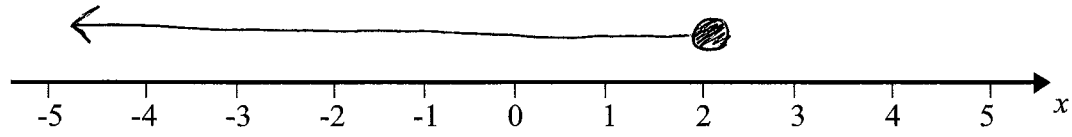
Show your answer on the number line.

$4x \leq x + 6$

$-x \qquad -x$

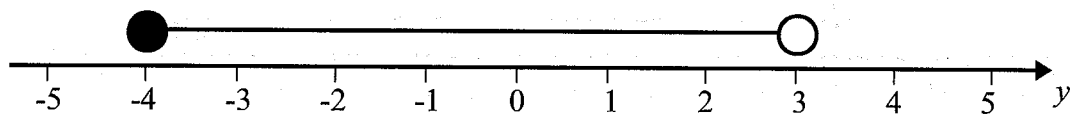
$3x \leq 6$

$x \leq 2$



(Total for question 7 is 3 marks)

8 Write down the inequality shown on the number line.

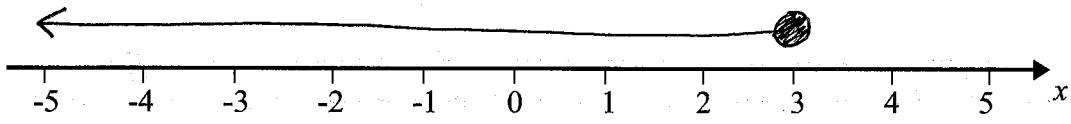


$-4 \leq y < 3$

(Total for question 8 is 2 marks)

9 (a) On the number line, show the inequality $x + 1 \leq 4$

$x \leq 3$



(2)

$5 < 2y < 12$ where y is an integer.

(b) Write down all the possible values of y.

$2.5 < y < 6$

3, 4, 5

(2)

(c) Solve $4 > 19 - 3x$

$+3x \qquad +3x$

$3x + 4 > 19$

$-4 \qquad -4$

$3x > 15$

$x > 5$

$x > 5$

(2)

(Total for question 9 is 6 marks)

10 n is an integer such that $-8 < 3n < 10$
Write down all the possible values of n.

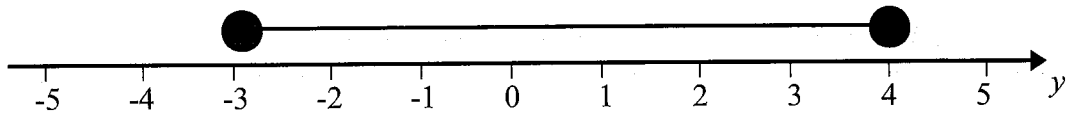
$\frac{-8}{3} < n < \frac{10}{3}$

$-2.\dot{6} < n < 3.\dot{3}$

$-2, -1, 0, 1, 2, 3$

(Total for question 10 is 2 marks)

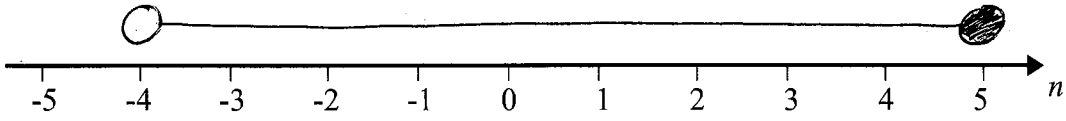
11 Write down the inequality shown on the number line.



$-3 \leq y \leq 4$

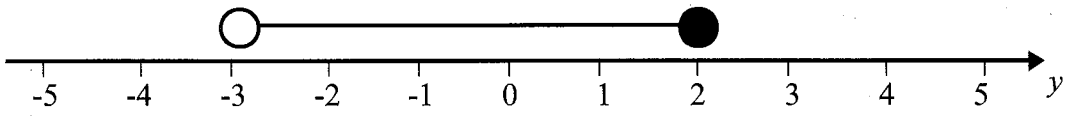
(Total for question 11 is 2 marks)

12 (a) On the number line, show the inequality $-4 < n \leq 5$



(2)

(b) Write down the inequality shown on the number line.



$-3 < y \leq 2$

(2)

(Total for question 12 is 4 marks)

13 Solve $2(3n - 5) > 12$

$6n - 10 > 12$

$+10 \quad +10$

$6n > 22$

$n > \frac{22}{6}$

$n > \frac{11}{3}$

$n > \frac{11}{3}$

(Total for question 13 is 2 marks)

14 n is an integer such that $-3 < 2n < 6$
Write down all the possible values of n.

$\frac{-3}{2} < n < 3$

$-1.5 < n < 3$

$-1, 0, 1, 2$

(Total for question 14 is 2 marks)

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15 Solve $3(n + 1) < 24$

$3n + 3 < 24$
$\quad -3 \quad -3$

$3n < 21$

$n < 7$

$n < 7$

(Total for question 15 is 2 marks)

---

16 Solve $4(2x + 1) > 9$

$8x + 4 > 9$

$8x > 5$

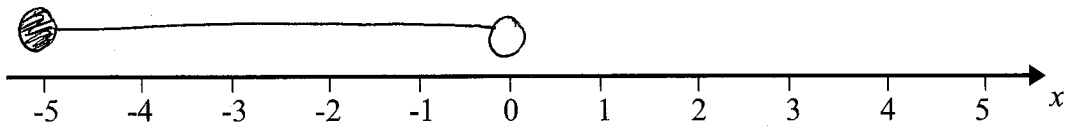
$x > \frac{5}{8}$

$x > \frac{5}{8}$

(Total for question 16 is 2 marks)

17 (a) On the number line, show the inequality $-3 \leq x + 2 < 2$

$-5 \leq x < 0$



(3)

$1 \leq 2y - 3 < 9$ where y is an integer.

$+3 \quad +3 \quad +3$

(b) Write down all the possible values of y.

$4 \leq 2y < 12$

$2 \leq y < 6$

2, 3, 4, 5

(3)

(c) Solve $4x - 4 \leq 7x - 19$

$-4x \quad -4x$

$-4 \leq 3x - 19$

$+19 \quad +19$

$15 \leq 3x$

$5 \leq x$

$x \geq 5$

$x \geq 5$

(3)

(Total for question 17 is 9 marks)